EXERCISE

Easy-to-Read Text

In this exercise, you will rank five versions of proposal text according to how easy they are to read.

1. Scan the five selections below.

2. With your team, agree on ranking from easy to hard.

EASY ←				→ HARD

3. List characteristics below that make the easiest easy to read.

_____ _____
_____ _____
_____ _____
_____ _____

Version 1

In all the experiments in the present work, it was found that the distributions were time-independent (this has been noted by most previous workers on liquid distribution) and this tends to indicate the existence of a stable flow pattern in the packed bed. It is thought to be of interest to determine to what extent this pattern of flow is dependent on bed structure and to what extent it is dependent on movement of a purely random nature of the initial liquid particles producing wetted paths through the packing. The effect on any given arrangement of packing of complete overload of the bed by high liquid flow rate (pre-flooding) might be predicted to be the alteration of a flow pattern, dependent upon which of the paths through the packing experienced wetting, whereas the effect of repacking or stirring the bed will be changes in the structure of the bed. Liquid spread measurements utilizing salt tracer techniques were carried out both prior to and subsequent to either pre-flooding of the packed bed or effecting changes in the structure of the bed by stirring (1/2 in. rings were used), other conditions, packed height and liquid rate, being kept constant. It was apparent from the results that both pre-flooding and bed stirring exerted a considerable influence on the stability of the flow pattern but that repacking exerted a substantially higher degree of influence than pre-flooding, hence indicating that bed structure is the factor of greatest importance in the determination of flow pattern stability.

Version 2

All our experiments showed that the distributions did not depend on time. This has been shown by most previous work on liquid distribution, and it suggests that the flow pattern in the packed bed is stable.

We thought it would be interesting to know to what extent this flow pattern depends on bed structure and to what extent it depends on the random movement of the initial liquid particles making wetted paths through the packing. In any given packing arrangement, repacking or "stirring" might be expected to change the structure of the bed; complete overloading of the bed with a high liquid flow-rate (pre-flooding) might be expected to alter a flow pattern depending on which paths through the pattern were wetted.

So we used salt tracer methods to measure liquid spread before and after stirring (using 1/2 in. rings) and before and after pre-flooding. Other conditions—packed height and liquid rate—were kept constant. Both stirring and pre-flooding affected the stable flow pattern, but stirring had much greater effect than pre-flooding; obviously, the structure of the bed is mainly responsible for the stable flow pattern.

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EXERCISE

Version 3

In all our experiments we found that our distributions were not dependent on time. This has been noted by the majority of our predecessors in work in the liquid distribution field, and we believe that this indicates the existence of a stable flow pattern in the packed bed.

We felt it would be of considerable interest if we were able to demonstrate in our experiments the extent to which this pattern depends on bed structure and to what extent it depends on a purely random movement of the initial liquid particles producing wetted paths through the packing. We might reasonably hypothesize that for any given packing arrangement, complete overloading of the bed by a high liquid flow-rate (known as pre-flooding) would alter our flow pattern, depending on which paths through the packing were wetted. We would expect, on the other hand, that repacking or stirring the bed would bring about changes in the fundamental structure of the bed.

We undertook measurements of liquid spread before and following either pre-flooding the packed bed or changing the structure of the bed by stirring (which we effected by means of 1/2 in. rings). Other conditions, packed height and liquid rate, we maintained at a constant level. Salt tracer techniques were our choice for the experiments.

Our results reveal that both pre-flooding and stirring the bed exert a marked influence on the stable flow pattern; nonetheless, the effect of repacking is of far greater magnitude. We must conclude, therefore, that the stable flow pattern is determined mainly by the structure of the bed.

Version 4

Time-independence of distributions, indicative of stable flow pattern in the packed bed, occurred in all experiments, confirming most previous liquid distribution work. The degree of dependence of this pattern on bed structure and/or on production of wetted paths through the packing by randomly moving particles of the initial liquid is of interest, as in any specified packing arrangement, complete bed overload by high liquid flow-rate (pre-flooding) might be expected to result in alteration of a flow pattern, dependent on which paths through the packing were wetted, while changes in bed structure might result from repacking or "stirring" the bed. Liquid-spread measurements by salt tracer techniques were made before and after either packed-bed pre-flooding or structure-changing by stirring (utilizing 1/2 in. rings); other conditions, packed height and liquid rate, being maintained constant. The effect of repacking on flow pattern stability is far in excess of that of pre-flooding. Hence, flow pattern stability is determined mainly by bed structure.

Version 5

All our experiments showed that the distributions did not depend on time. This agrees with previous work on liquid distribution. Therefore, the liquid in the packed bed probably flows in a stable pattern.

We looked at the extent to which this flow pattern depends on:

1. Bed structure
2. Random movement of the initial liquid particles making wetted paths through the packing

We thought that repacking or stirring would change the bed structure. We also expected that saturation with a high liquid flow-rate (pre-flooding) would alter the flow pattern.

We measured liquid spread, before and after both stirring and pre-flooding using salt tracer methods. We kept the packed height and liquid flow rate constant.

Stirring and pre-flooding both altered the stable flow pattern. Stirring had the most effect because the structure of the bed mainly causes the stable flow pattern.

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